Department of Computer Science and Engineering (CSE)
BRAC University

Practice Problem Set 3.2

CSE251 – Electronic Devices and Circuits

DIODE CIRCUIT MODELS

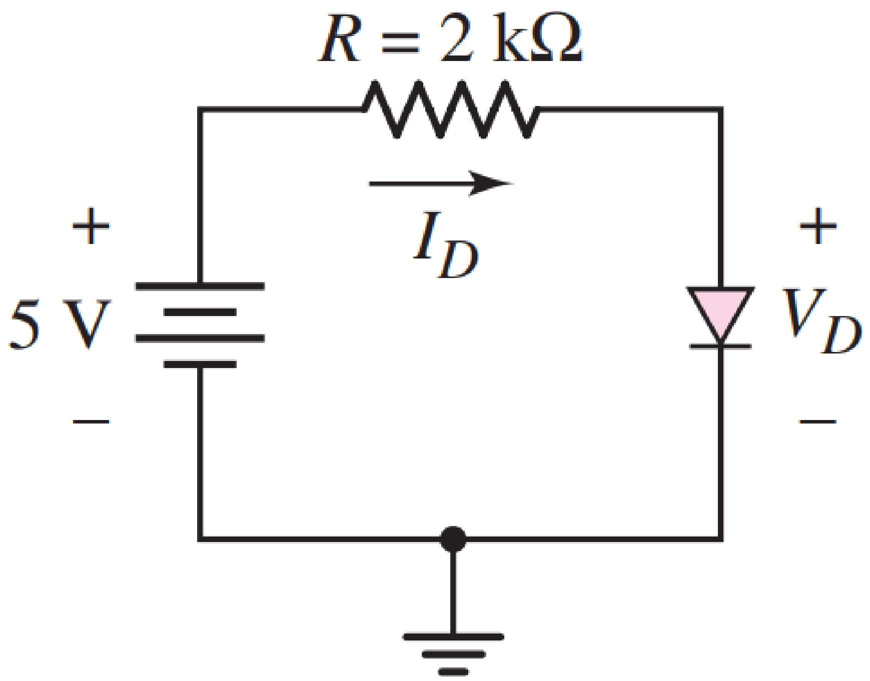
Method of assumed states, Exponential Model, and
Piecewise Linear Models: CVD+R and CVD

Course Description, COs, and Policies

Midterm and Final Questions

Problem 1

- If the diode in the following circuit has a reverser saturation current $I_o = 10^{-13} A$, follow an iterative approach and write a program to determine I_D and V_D. Assume the ideality factor $\eta = 1$ and the thermal voltage $V_T = 25\ mV$.



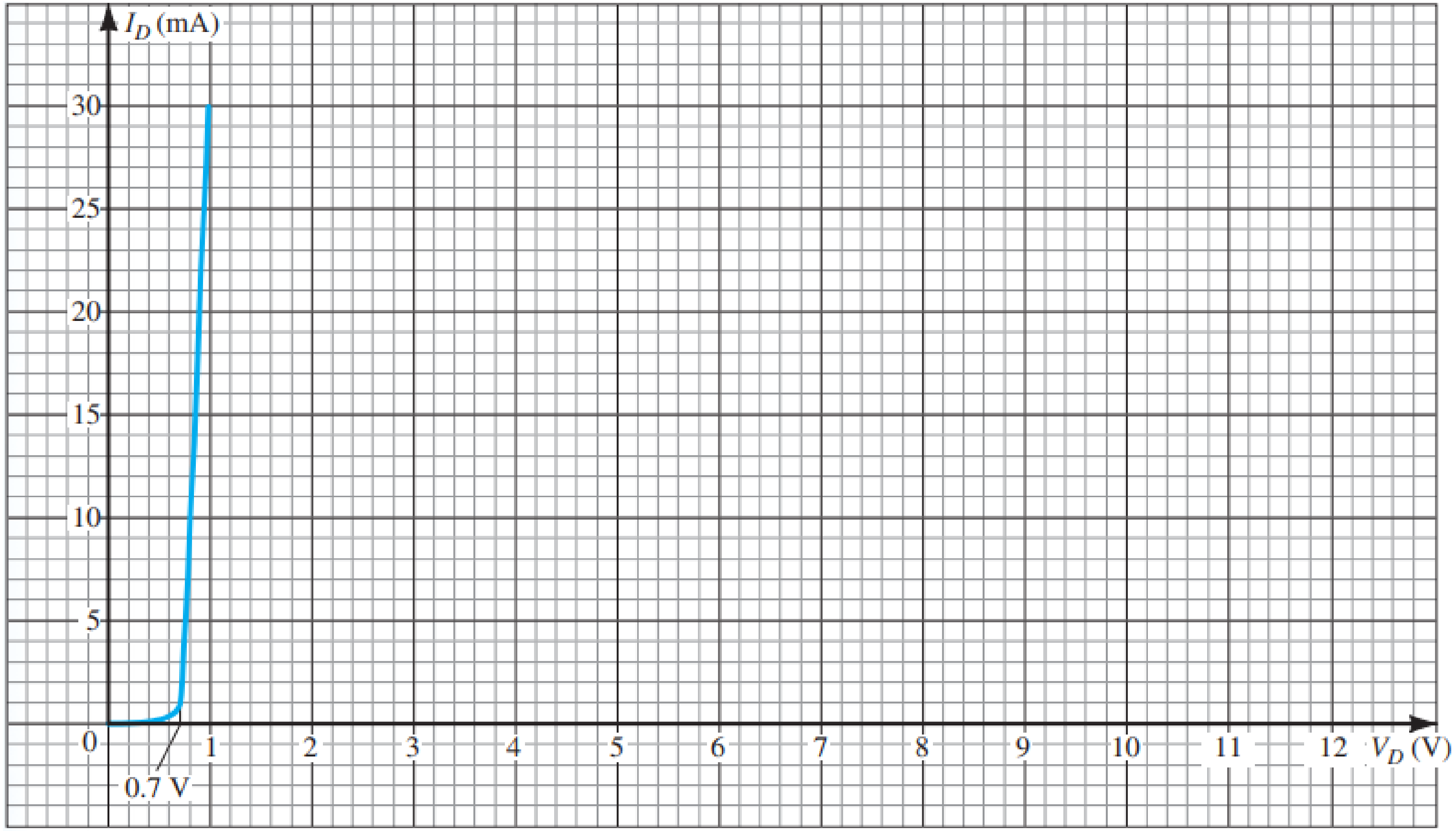
Hint: There will be two simultaneous equations: one is the diode's current-voltage relation and the other is the result of applying KVL around the loop.

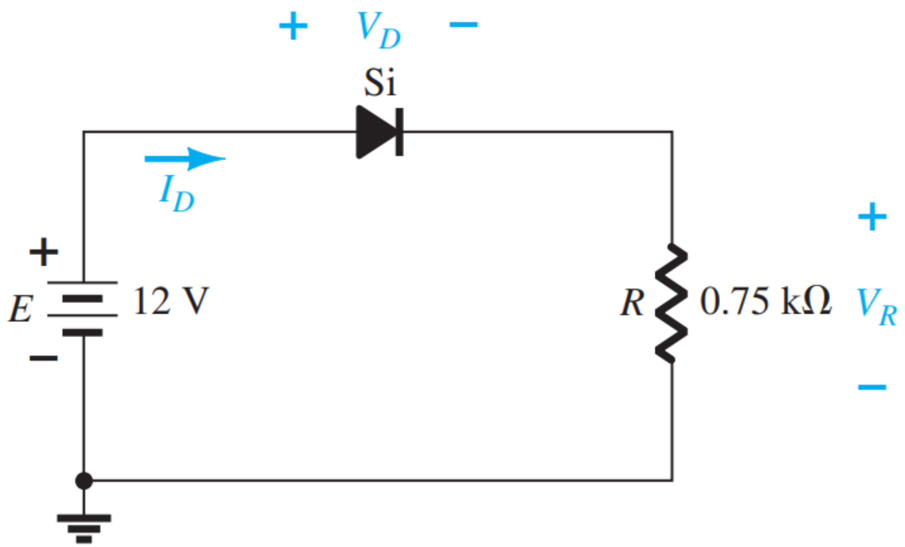
$I_D = I_o \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$ and $I_D = -\frac{1}{2} V_D + \frac{5}{2}$

Take an array of V_D with values in a reasonable range. For example, from $0.400\ V$ to $0.800\ V$. For each V_D, evaluate I_D using both the equations. The value of V_D for which both the equations of I_D yields the closest value is the answer.

Problem 2

- The $I - V$ characteristic of the diode in the following circuit has been plotted below. Determine I_D and V_D.

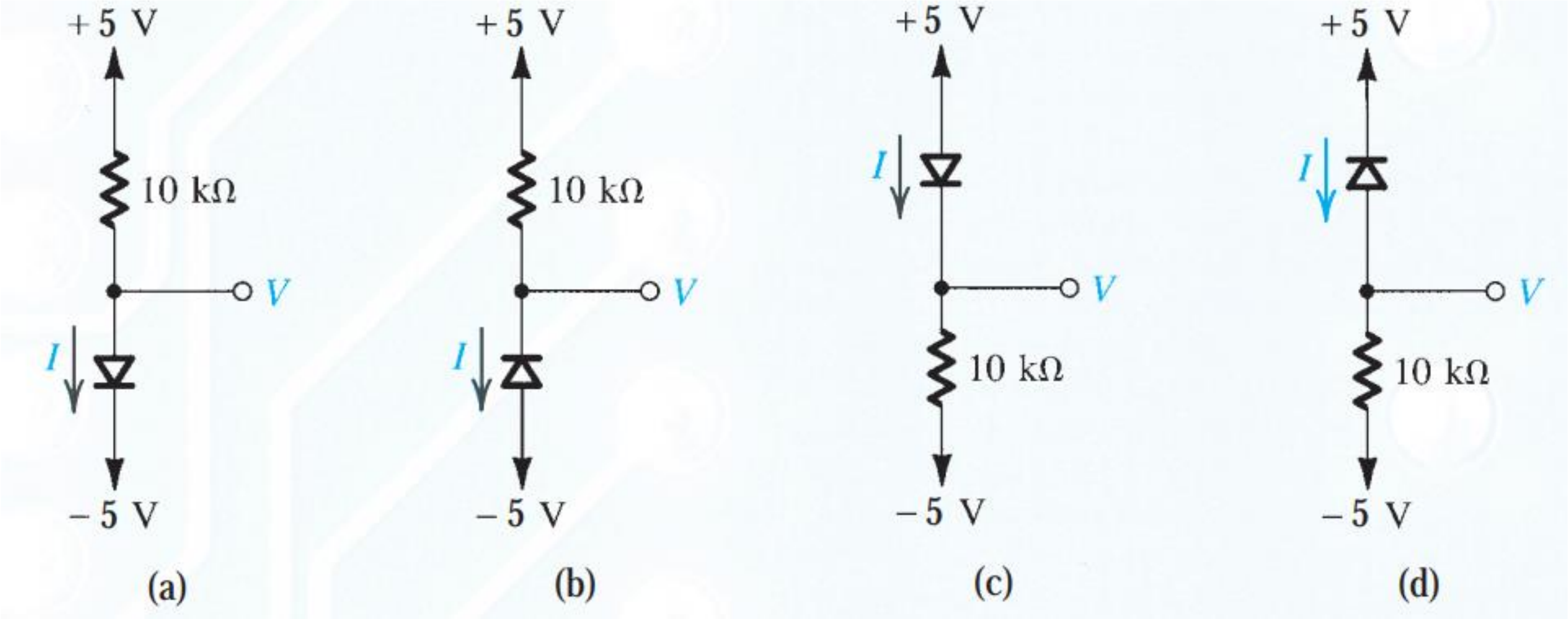




Ans: $I_D = 15\ mA, V_D = 0.82\ V$

Problem 3

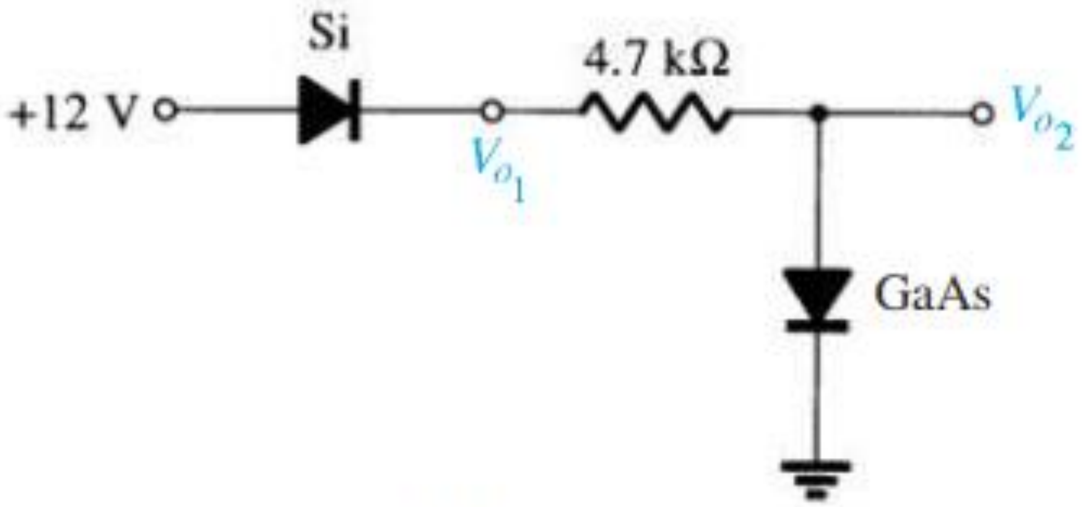
- For the circuits shown below, using ideal diode model, find the values of the voltages and currents indicated.



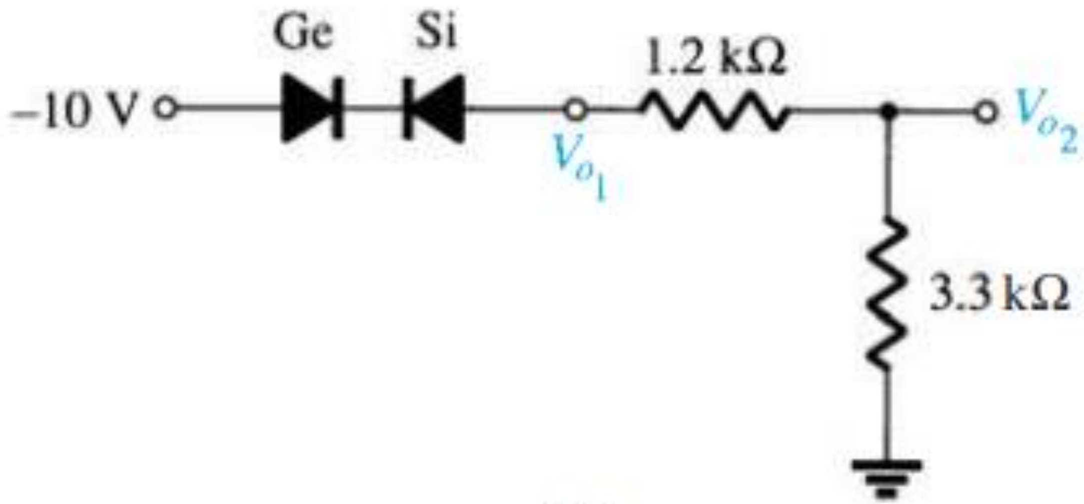
Ans: $(\boldsymbol{a})\ \boldsymbol{I = 1\ mA, V = -5\ V; (b)\ I = 0\ mA, V = 5\ V; (c)\ I = 1\ mA, V = 5\ V; (d)\ I = 0\ mA, V = -5\ V}$

Problem 4

- Determine V_{o_1} and V_{o_2} for the circuits shown below. Use $CVD + R$ Model with $V_{D_{o,\,Si}} = 0.7\,V$, $r_o = 20\,\Omega$, $V_{D_{o,\,Ge}} = 0.3\,V$, $r_o = 10\,\Omega$, and $V_{D_{o,\,GaAs}} = 1.2\,V$, $r_o = 5\,\Omega$.



(a)

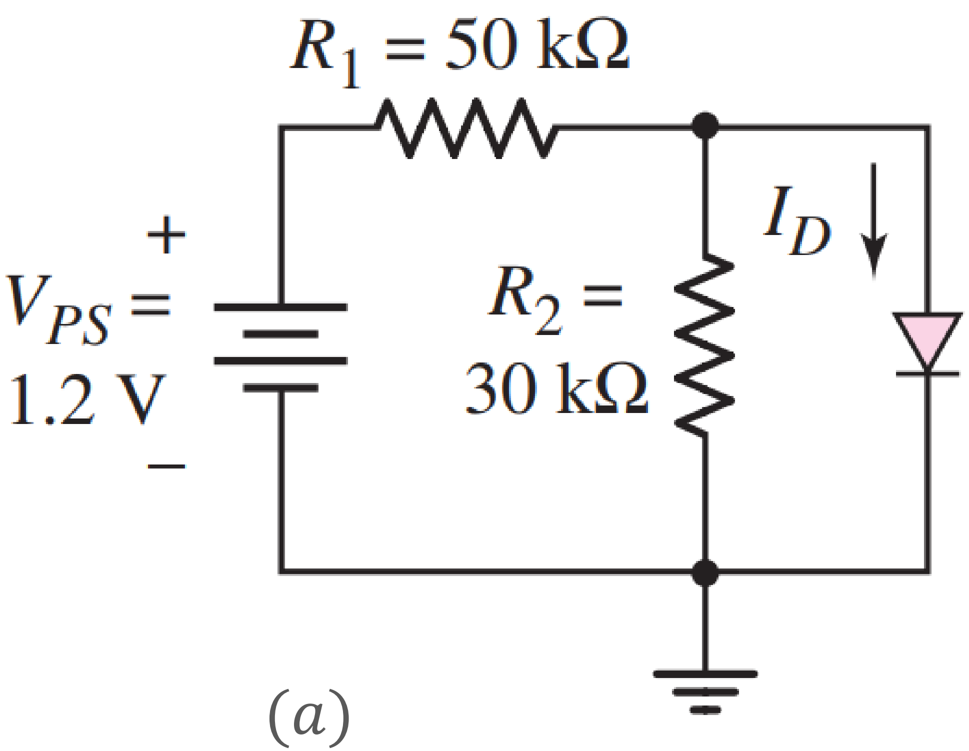


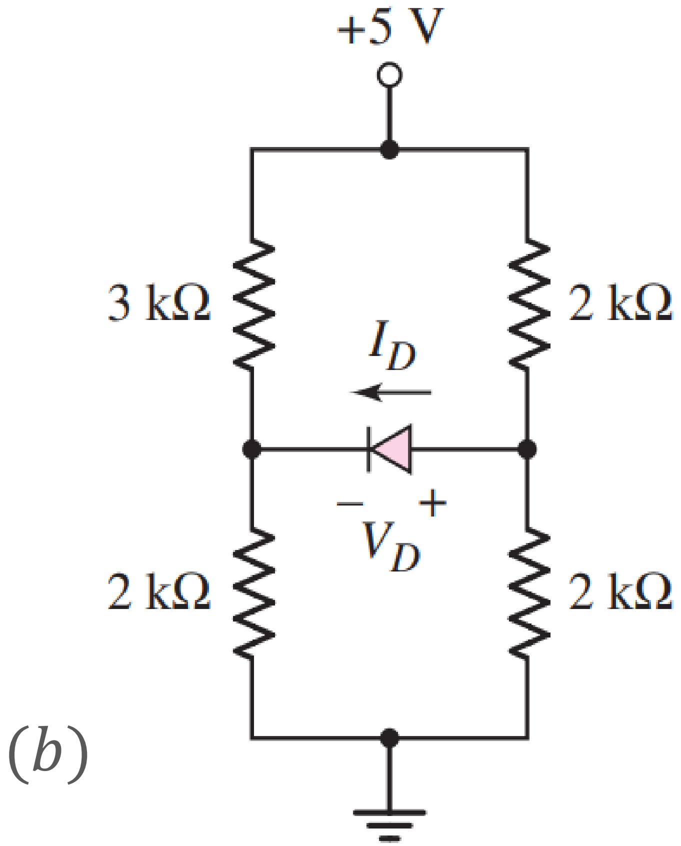
(b)

Ans: $(\boldsymbol{a})\ \boldsymbol{V_{o_1} = 11.3\ V, V_{o_2} = 1.21\ V;\ (b)\ V_{o_1} = 0\ V, V_{o_2} = 0\ V}$

Problem 5

- For the circuits shown below, using CVD model (with $V_{D_o} = 0.7\ V$), find I_D.

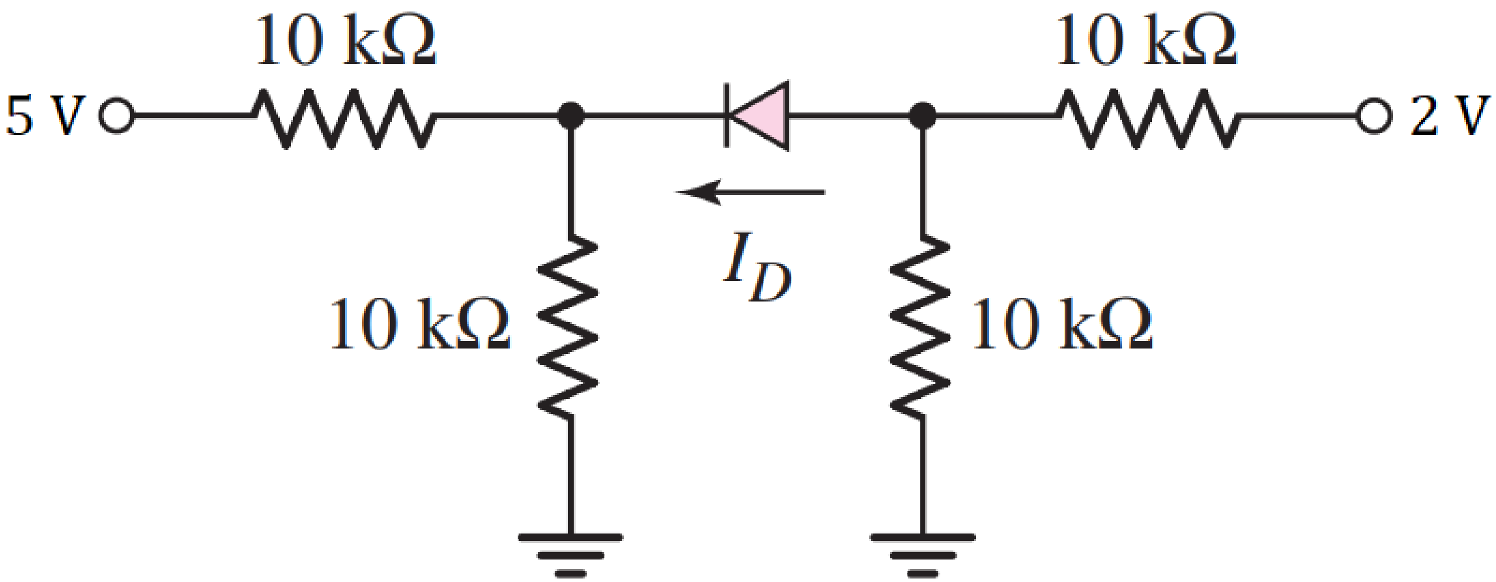




Ans: $(\boldsymbol{a})\ \boldsymbol{I_D = 0\ mA};\ (\boldsymbol{b})\ \boldsymbol{I_D = 0\ mA}$

Problem 6

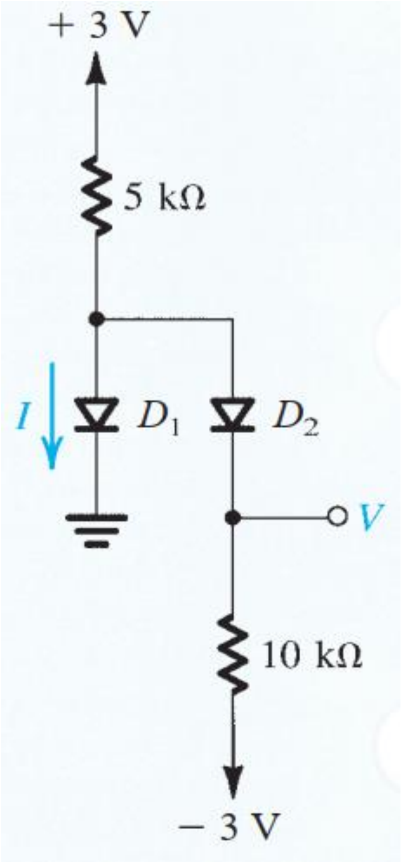
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine I_D.



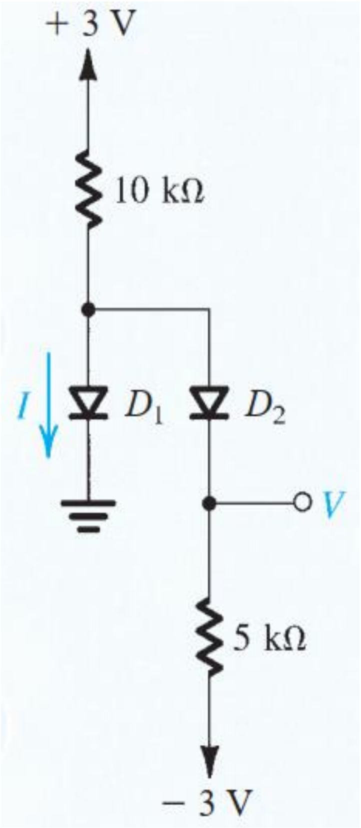
Ans: $\boldsymbol{I_D = 0\ mA}$

Problem 7

- For the circuits shown below, using ideal diode model, find the values of the voltages and currents indicated.



(a)

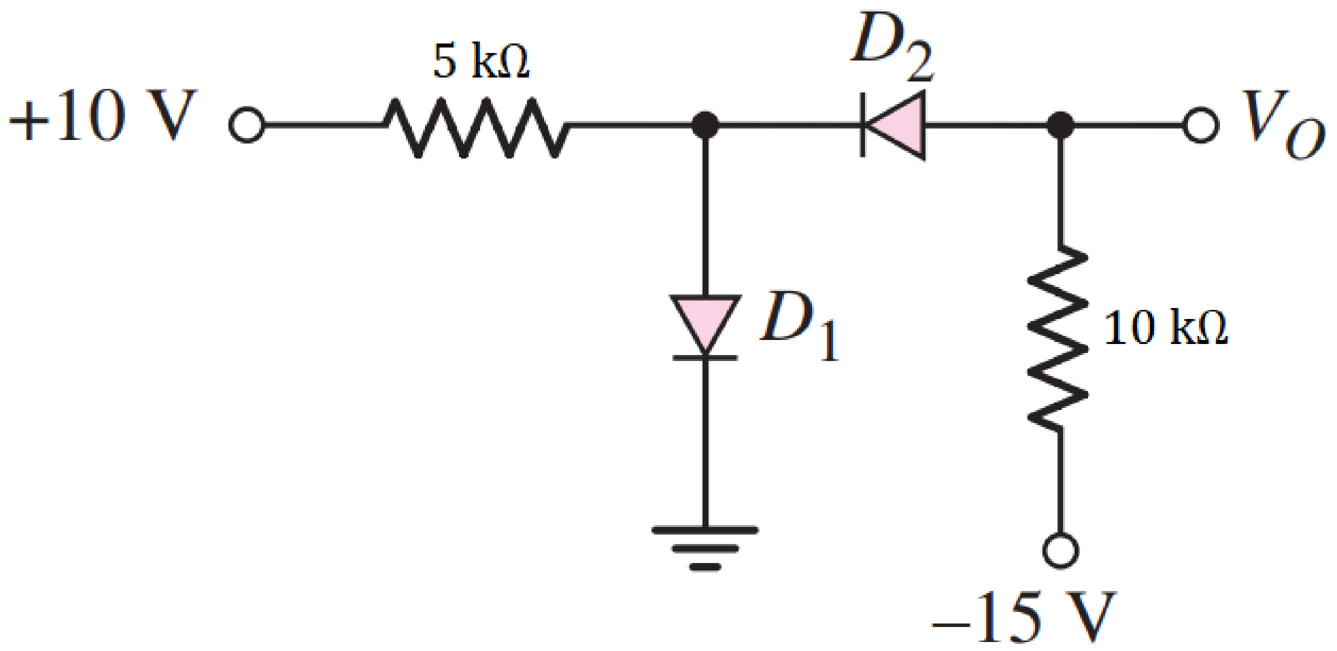


(b)

Ans: $(\boldsymbol{a})\ \boldsymbol{D_1\ ON, D_2\ ON, I = 0.3\ mA, V = 0\ V;}\ (\boldsymbol{b})\ \boldsymbol{D_1\ OFF, D_2\ ON, I = 0\ mA, V = -1\ V}$

Problem 8

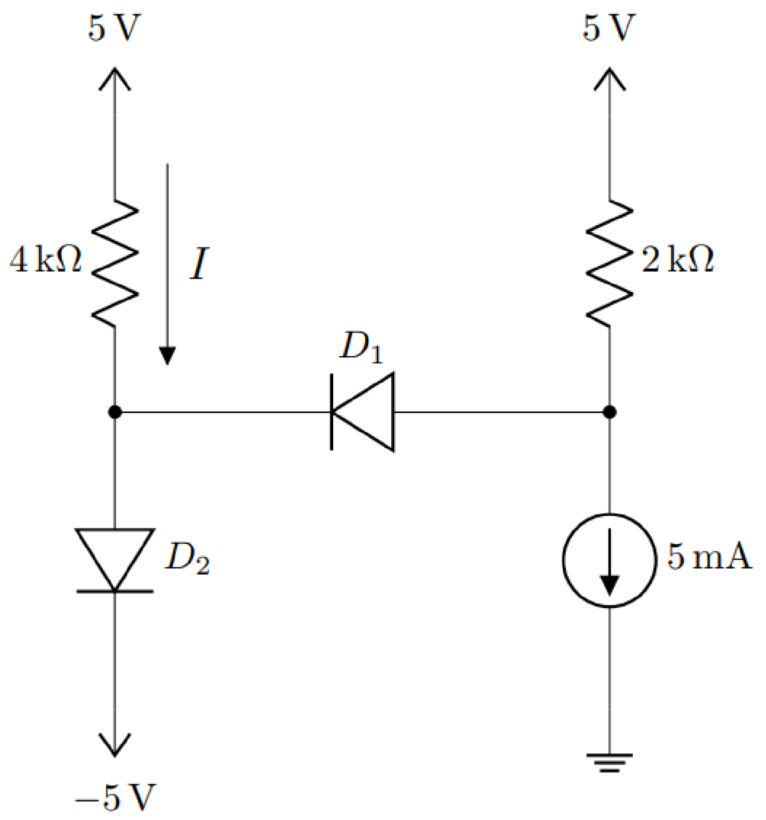
- For the circuit shown below, using CVD model (with $V_{D_O} = 0.7\ V$), determine V_O.



Ans: $\boldsymbol{D_1\ ON, D_2\ OFF, V_O = -15\ V}$

Problem 9

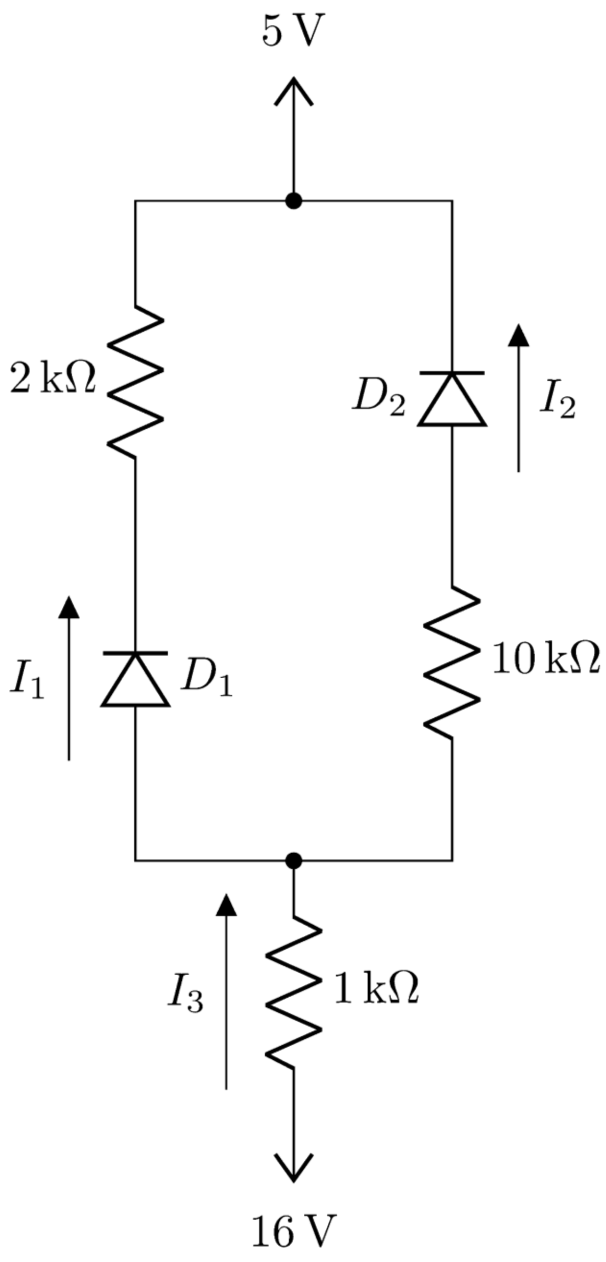
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine I.



Ans: $\boldsymbol{D_1\ OFF, D_2\ ON, I = 2.325\ mA}$

Problem 10

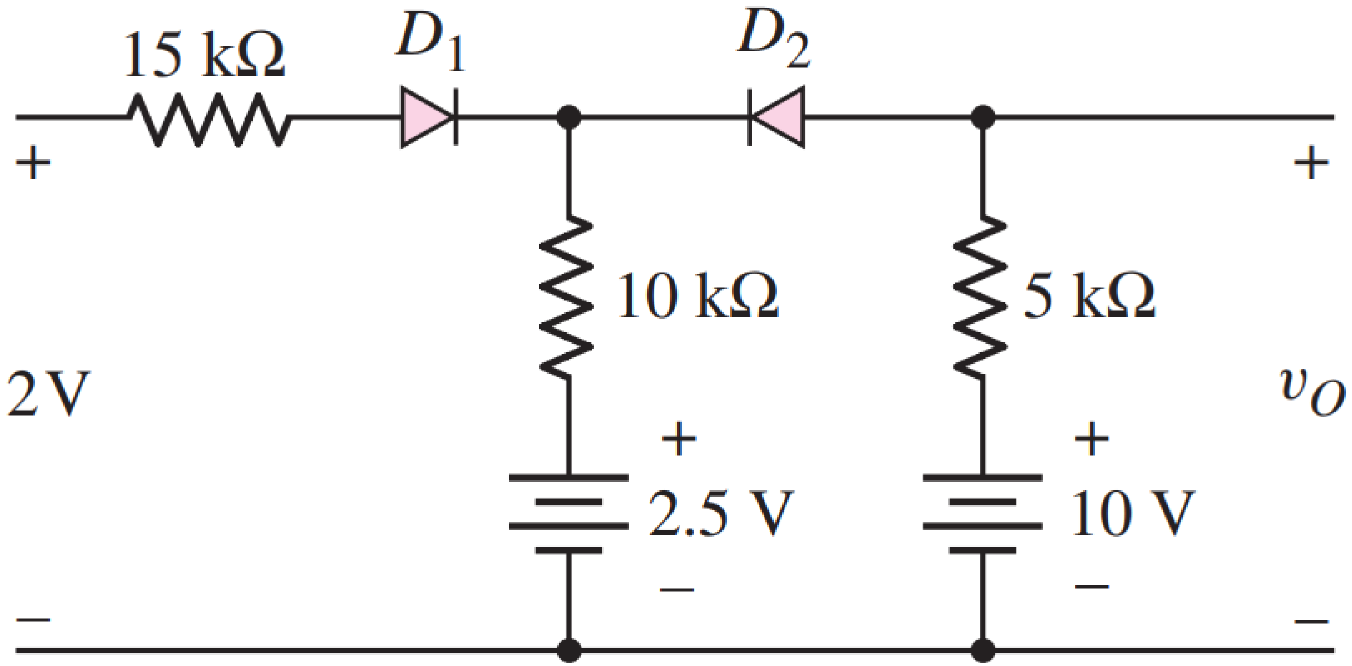
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine I_1, I_2, and I_3.



Ans: $\boldsymbol{D_1\ ON, D_2\ ON, I_1 = 3.22\ mA,\ \ I_2 = 0.64\ mA,\ \ I_3 = 3.86\ mA}$

Problem 11

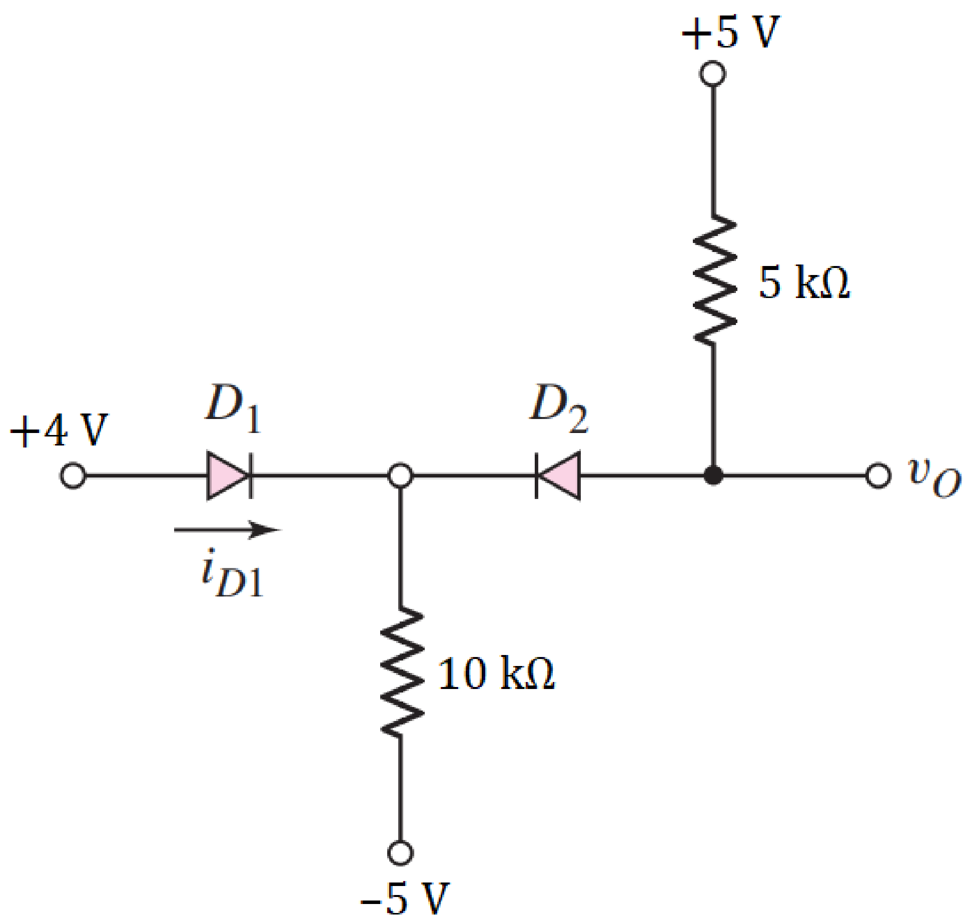
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine v_O.



Ans: $\boldsymbol{D_1\ OFF, D_2\ ON, v_O = 7.735\ V}$

Problem 12

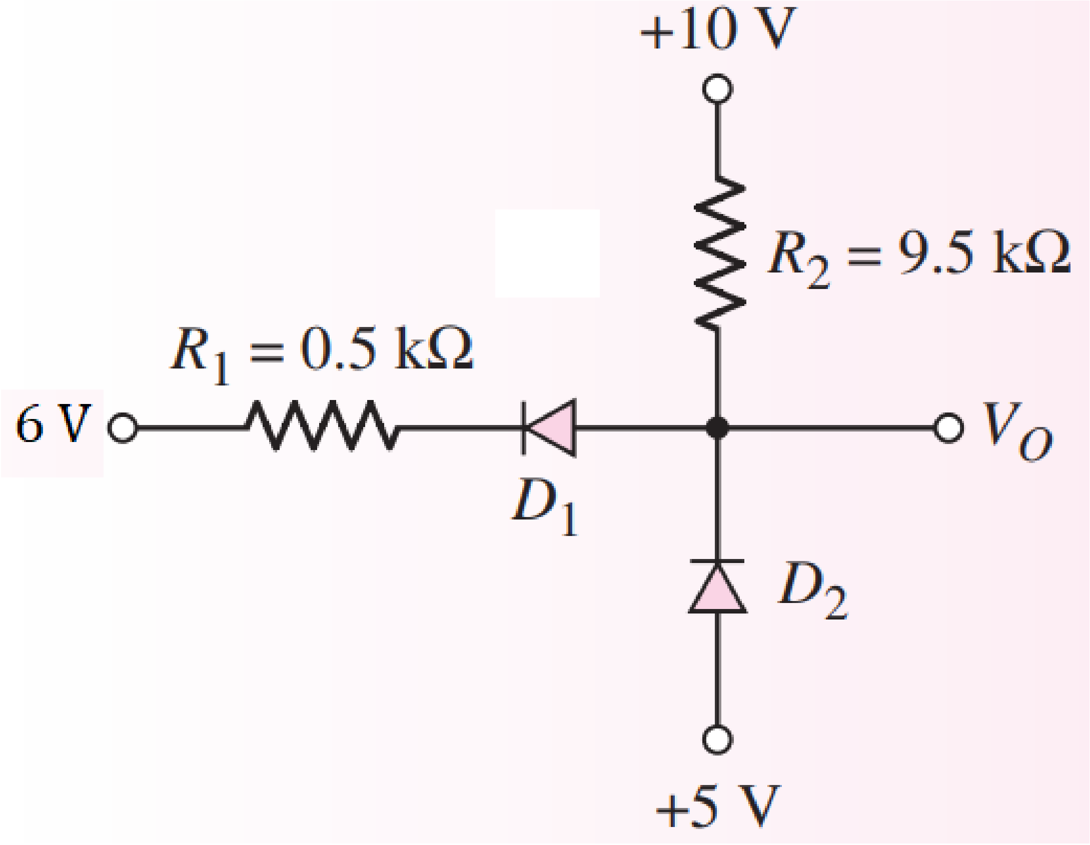
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine v_O.



Ans: $\boldsymbol{D_1\ ON, D_2\ ON, v_O = 4\ V}$

Problem 13

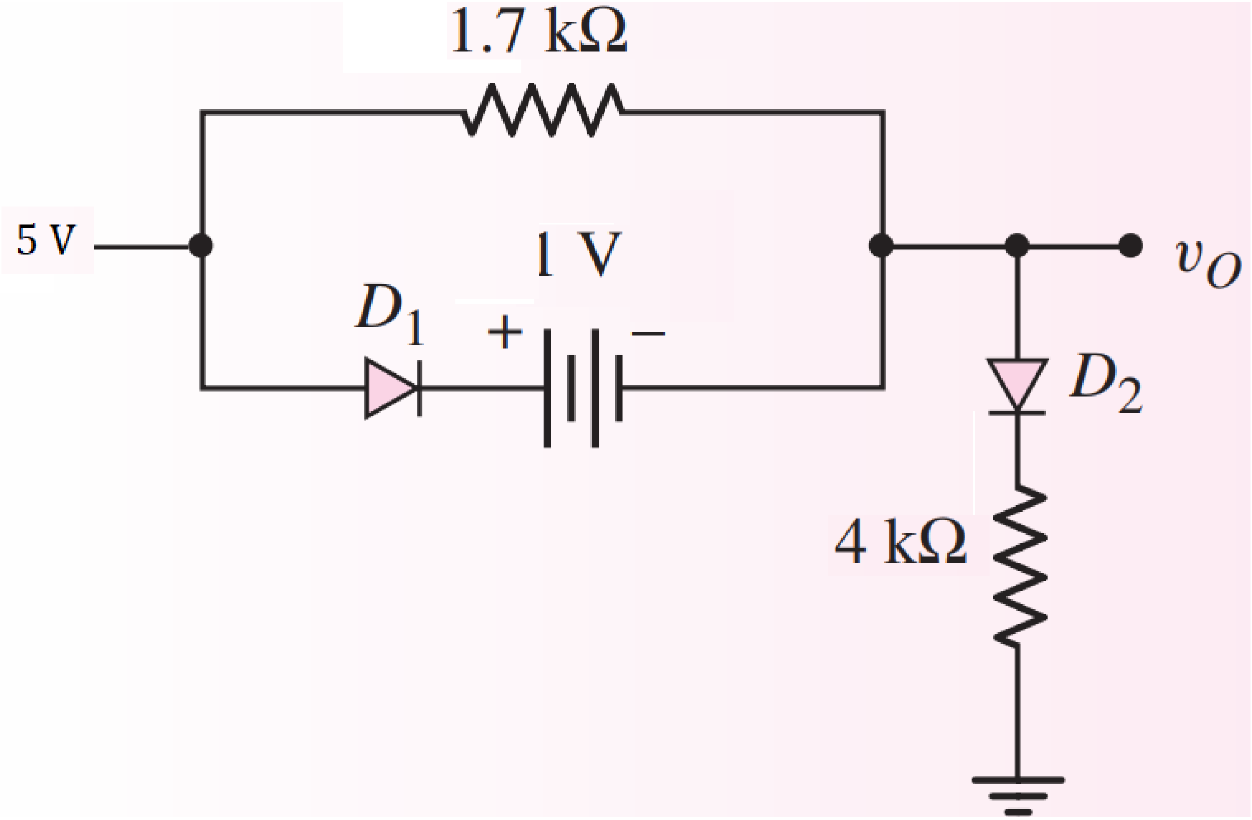
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine V_O.



Ans: $\boldsymbol{D_1\ ON, D_2\ OFF, V_O = 6.865\ V}$

Problem 14

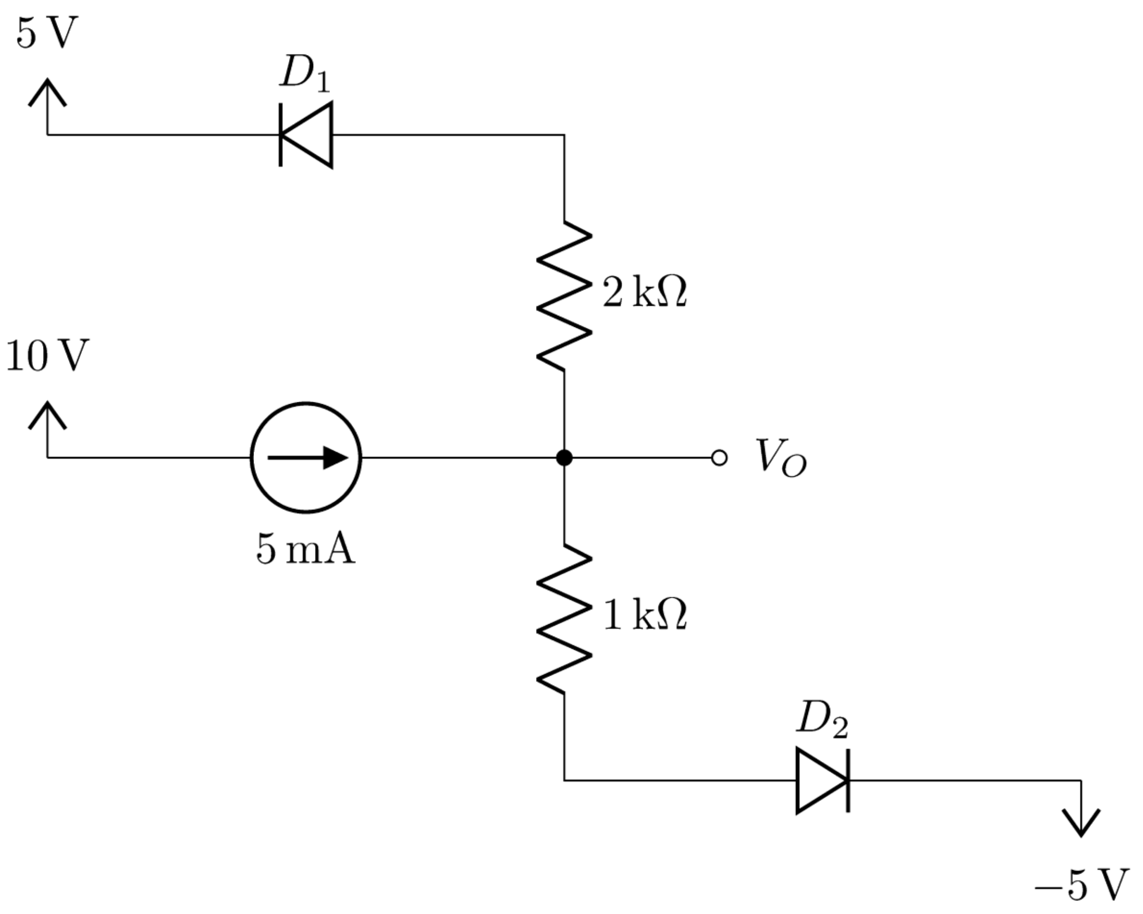
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine v_O.



Ans: $\boldsymbol{D_1\ OFF, D_2\ ON, v_O = 3.72\ V}$

Problem 15

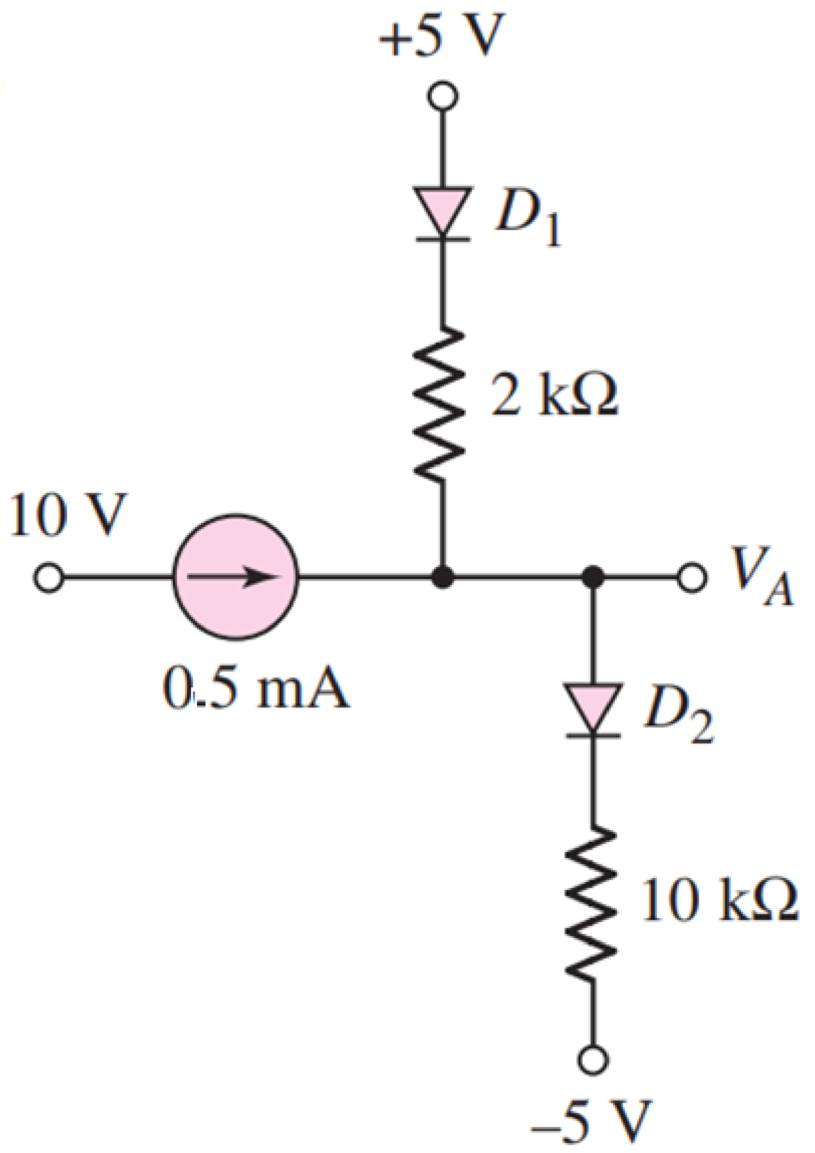
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine V_O.



Ans: $\boldsymbol{D_1\ OFF, D_2\ ON, V_O = 0.7\ V}$

Problem 16

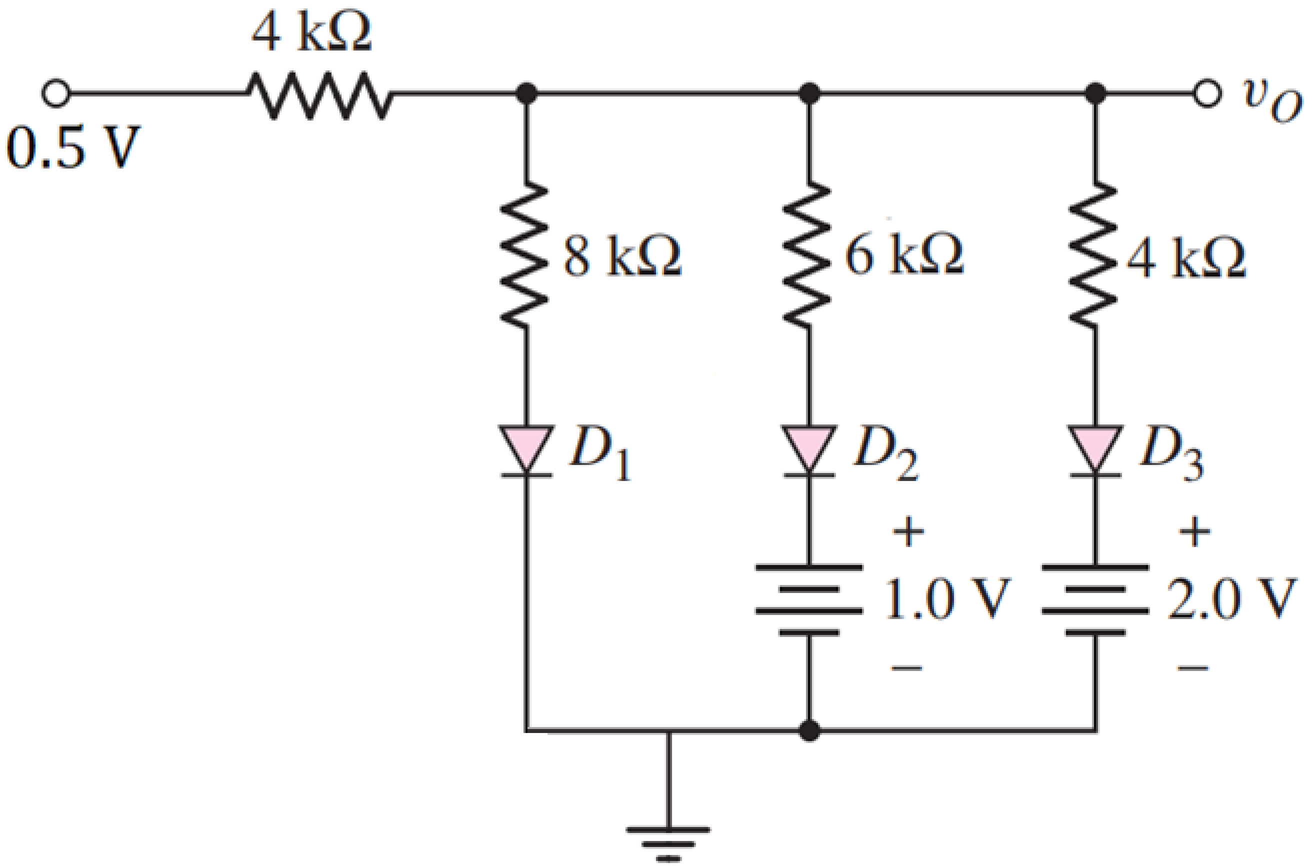
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine V_A.



Ans: $\boldsymbol{D_1\ ON, D_2\ ON, V_A = 3.7\ V}$

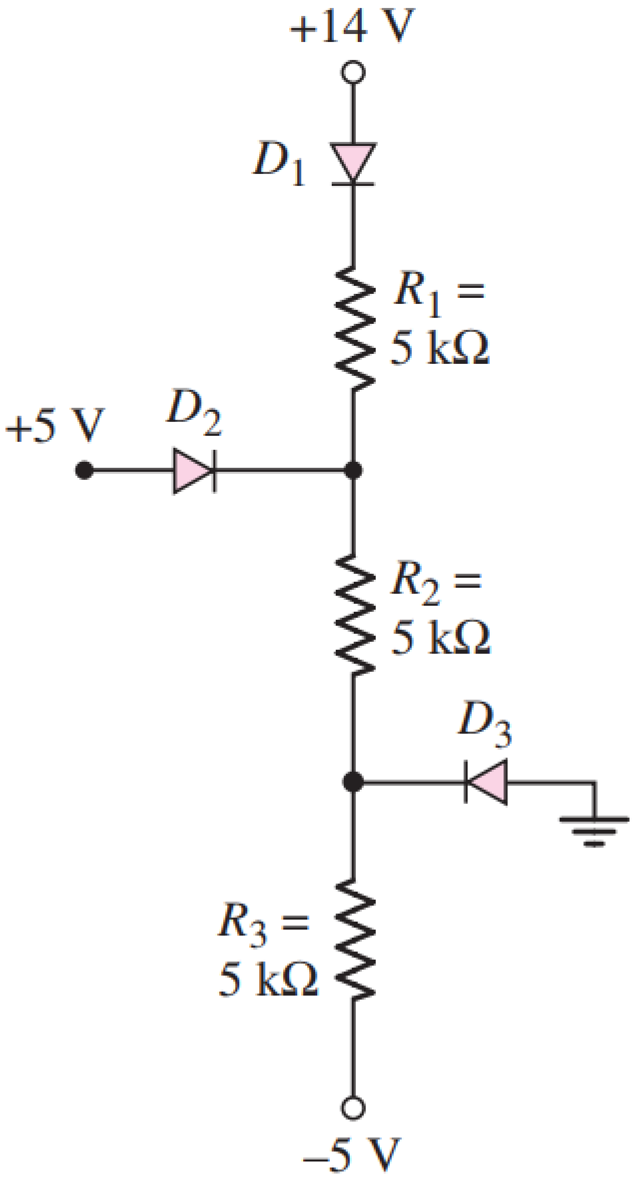
Problem 17

- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), show that all the diodes remain off.



Problem 18

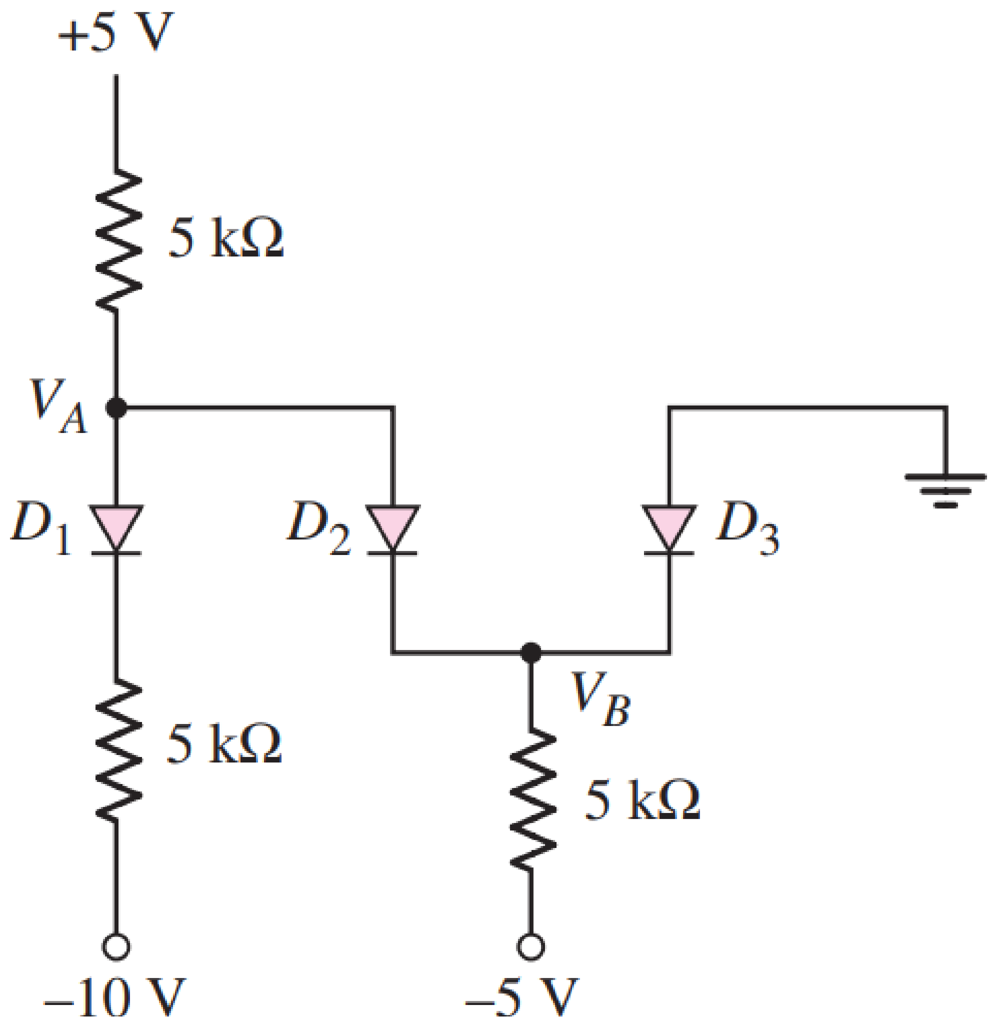
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine the voltage across R_2.



Ans: $\boldsymbol{D_1\ ON, D_2\ OFF, D_3\ OFF, V_{R_2} = 6.11\ V}$

Problem 19

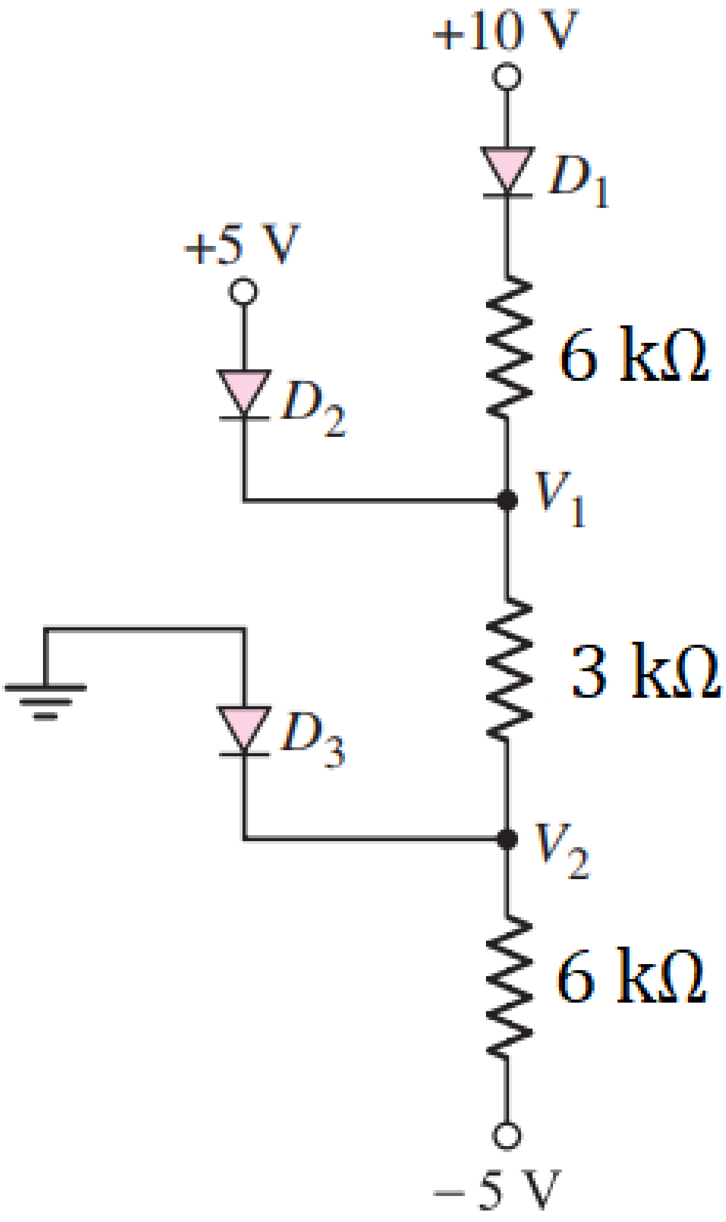
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\,V$), determine V_A and V_B.



Ans: $\boldsymbol{D_1\ ON, D_2\ OFF, D_3\ ON, V_A = -2.16\ V, V_B = -0.65\ V}$

Problem 20

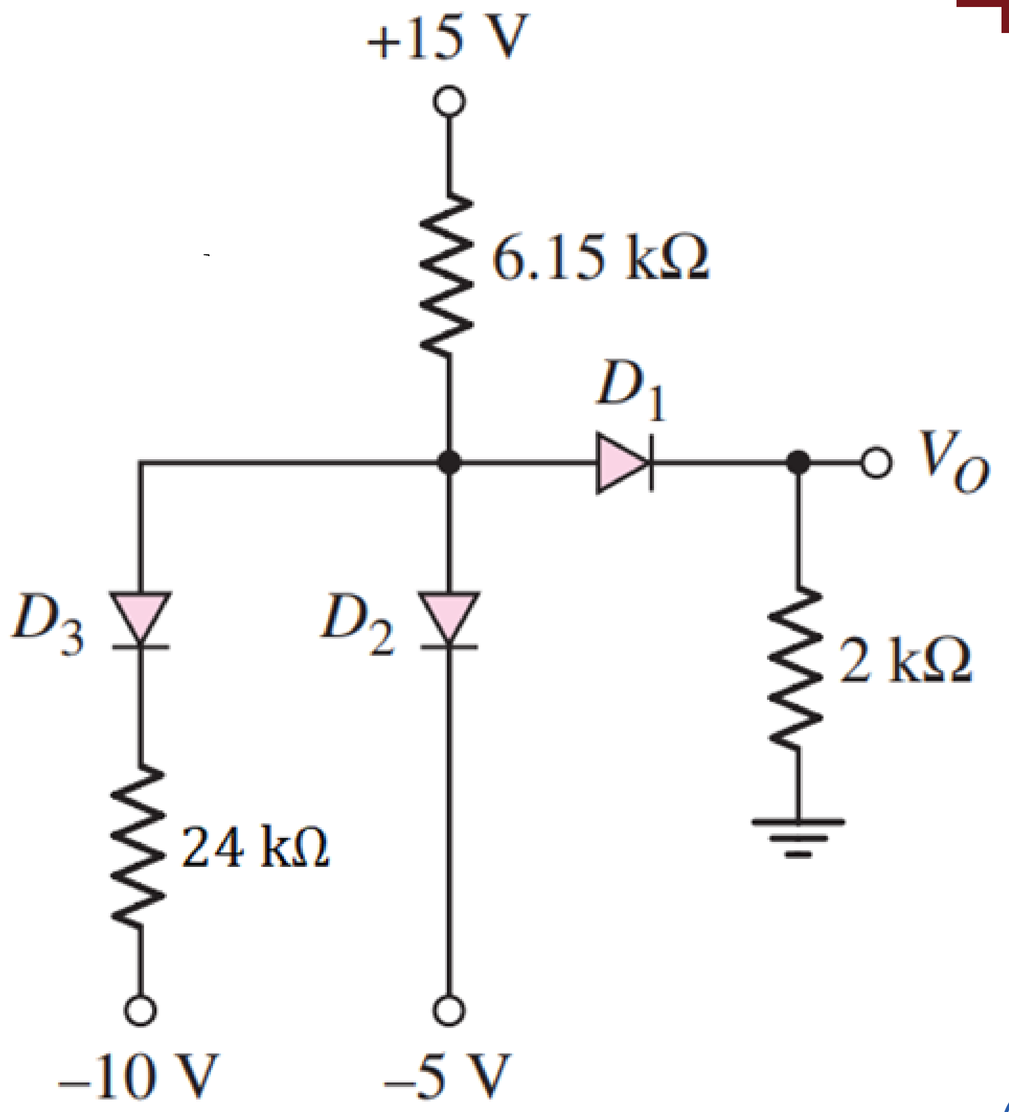
- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine V_1 and V_2.



Ans: $\boldsymbol{D_1\ ON, D_2\ ON, D_3\ OFF, V_1 = 4.38\ V, V_2 = 1.23\ V}$

Problem 21

- For the circuit shown below, using CVD model (with $V_{D_o} = 0.7\ V$), determine V_O.



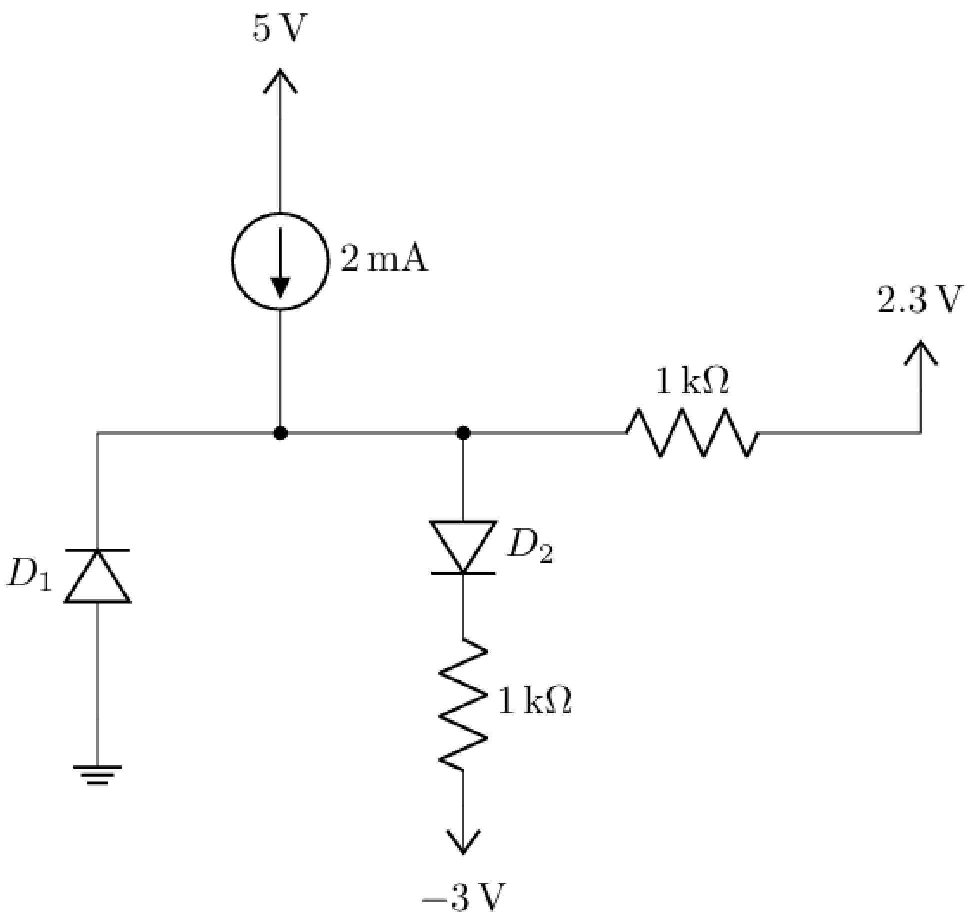
Ans: $\boldsymbol{D_1\ OFF, D_2\ ON, D_3\ ON, V_O = 0\ V}$

Up Next

Relevant Questions from Past Semesters

Mid – Fall 2025

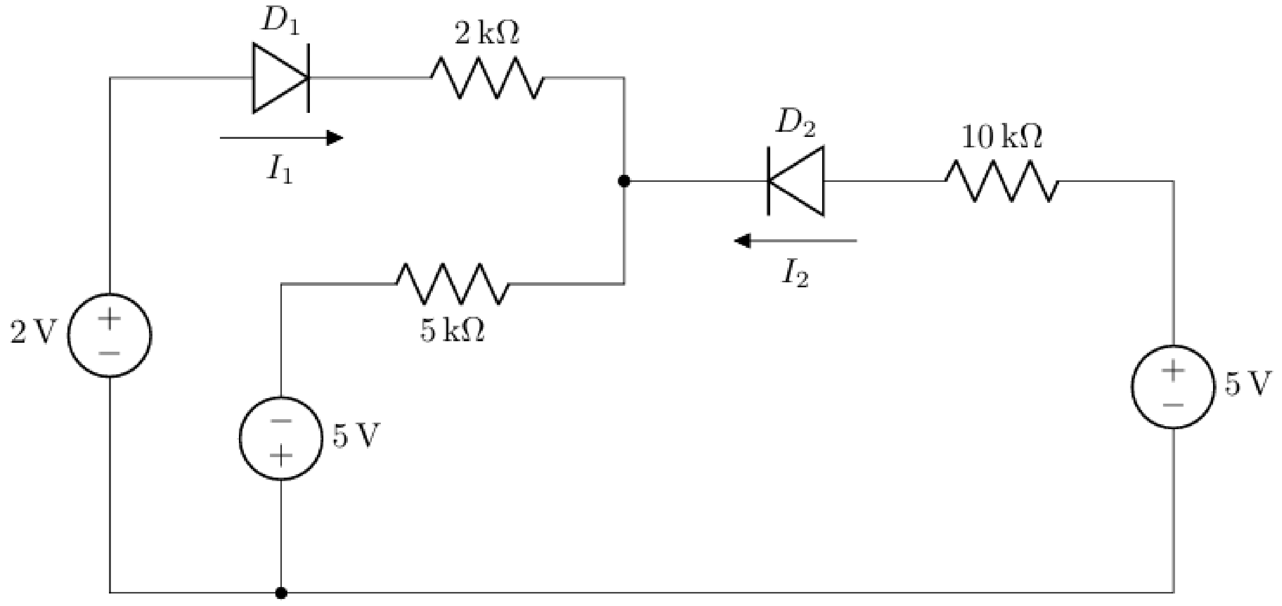
- Assuming a cut-in voltage of $0.7\ V$ for the identical diodes D_1 and D_2 in the circuit shown below. Analyze the circuit to determine which diode(s) are conducting. Additionally, calculate the power dissipated by any conducting diode(s).



Ans: $\boldsymbol{D_1 \rightarrow OFF, V_{D_1} = -1\ V;\ D_2 \rightarrow ON, I_{D_2} = 3.3\ mA, P_{D_2} = 2.31\ mW}$

Mid – Summer 2025

- Assuming a cut-in voltage of $0.7\,V$ for the identical diodes D_1 and D_2 in the circuit shown below. Analyze the circuit to determine which diode(s) are conducting. Additionally, calculate the power dissipated by each diode.

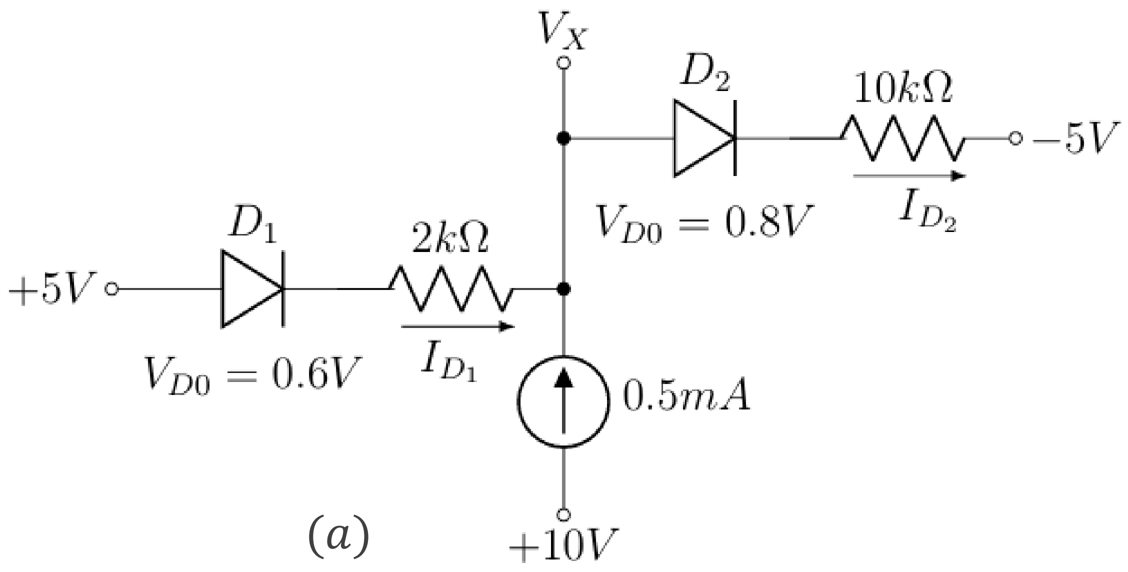


Ans: $\boldsymbol{D_1 \rightarrow ON, I_{D_1} = 0.6\,mA, P_{D_1} = 0.42\,mW;\ D_2 \rightarrow ON,\ I_{D_2} = 0.42\,mA; P_{D_2} = 0.294\,mW}$

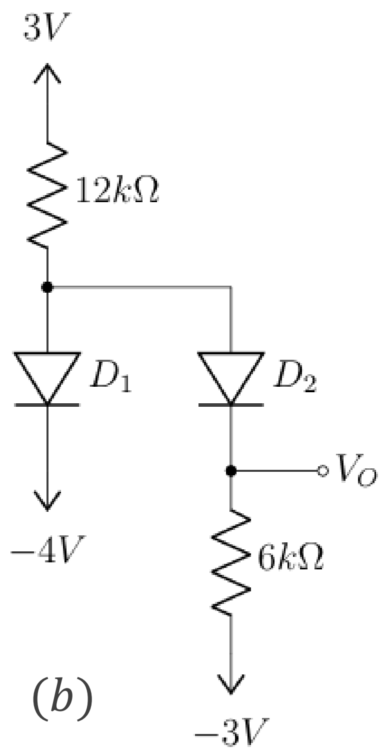
Mid – Spring 2025

a. Analyze the circuits shown below and calculate I_{D_1} and I_{D_2}.

b. Assuming a cut-in voltage of $0.7\ V$ for the identical diodes D_1 and D_2, calculate V_O.



(a)

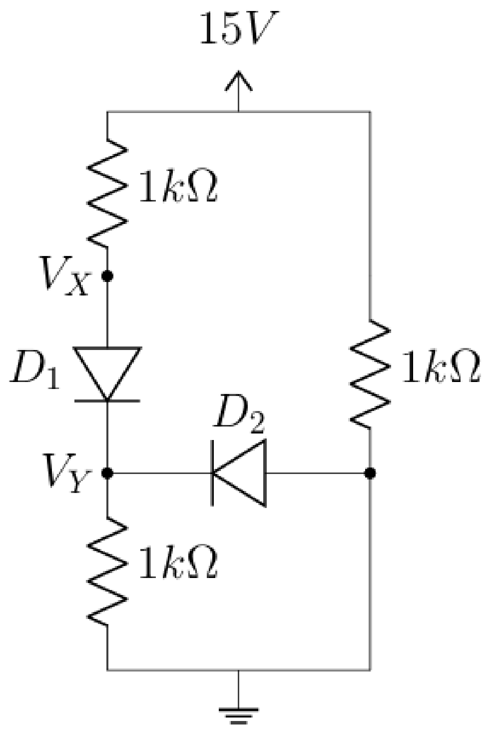


(b)

Ans: (a) $D_1 \rightarrow ON, I_{D_1} = 0.3\ mA;\ D_2 \rightarrow ON,\ I_{D_2} = 0.8\ mA$
(b) $D_1 \rightarrow ON, I_{D_1} = 0.525\ mA;\ D_2 \rightarrow OFF, V_{D_2} = -0.3\ V; V_O = -3\ V$

Mid – Fall 2024

- Analyze the following circuit, and calculate V_X, V_Y , I_{D_1}, and I_{D_2} using the method of assumed states. You must validate your assumptions. Use a cut-in voltage of $V_{D_0} = 0.7\ V$ for both diodes.



Ans: $\boldsymbol{D_1 \rightarrow ON, I_{D_1} = 7.15\ mA;\ D_2 \rightarrow OFF,\ V_{D_2} = -7.15\ V}$

Mid – Summer 2024

a. The diodes D_1 and D_2 in *Figure – 1* have cut-in voltages of $0.5\,V$ and $0.6\,V$, respectively. Determine V_O.

b. Assuming a cut-in voltage of $0.7\,V$ for the identical diodes D_1 and D_2 in the circuits shown below determine I_{D_1} and I_{D_2} from the circuit in *Figure – 2* and I_{ab} from the circuit in *Figure – 3*.

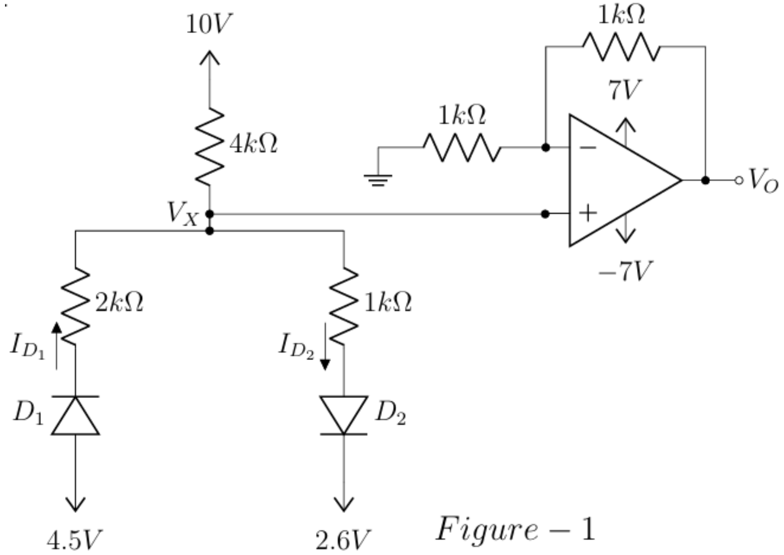


Figure – 1

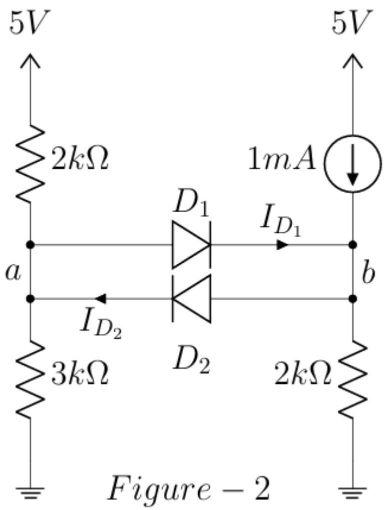


Figure – 2

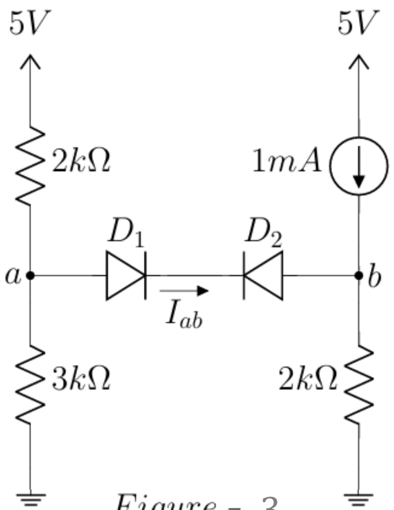
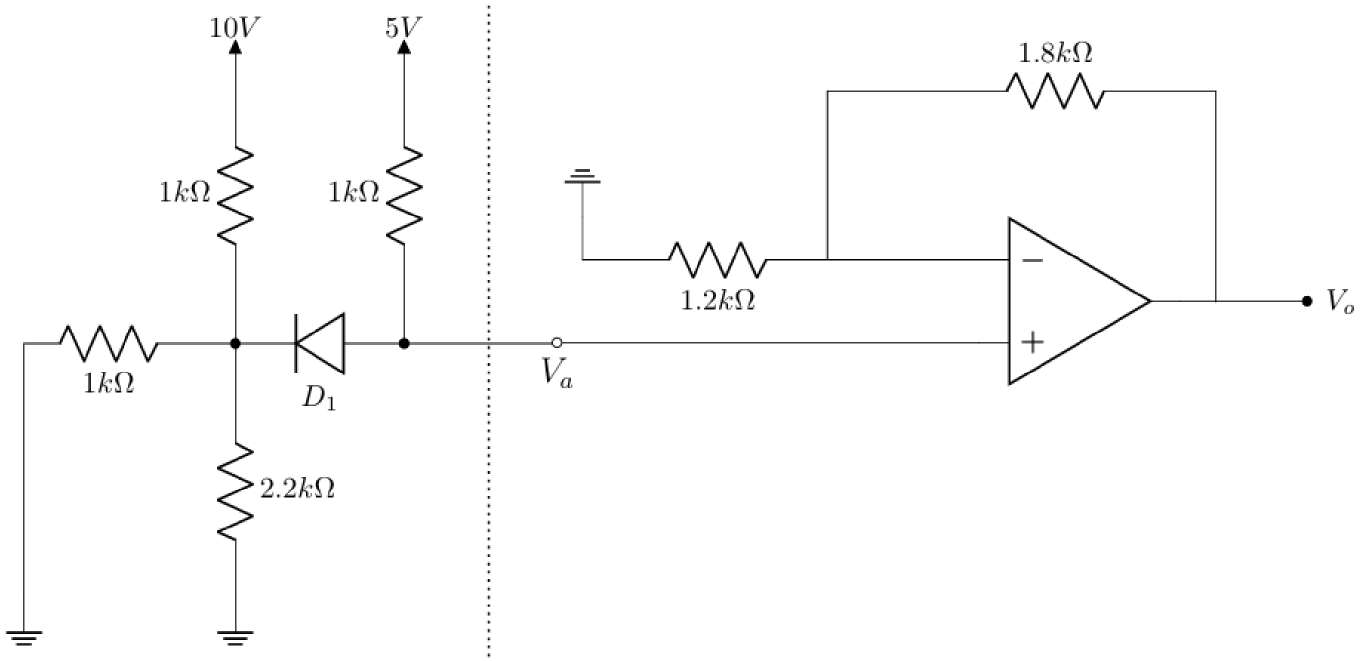


Figure - 3

Ans: Figure – 1: $D_1 \rightarrow OFF, D_2 \rightarrow ON, I_{D_2} = 1.36\,mA, V_x = 4.56\,V,\ V_O = 7\,V$
Figure – 2: $D_1 \rightarrow ON, I_{D_1} = 0.088\,mA$; $D_2 \rightarrow OFF,\ V_{D_2} = -0.7\,V$; Figure – 3: $D_1 \rightarrow OFF, D_2 \rightarrow OFF, I_{ab} = 0\,mA$

Mid – Summer 2023

- The saturation voltages for the op-amp in the following circuit are $V_{Sat}^{+} = +10\ V$, $V_{Sat}^{-} = -10\ V$, and the cut-in voltage for the diode D_1 is $V_{D_0} = 0.7\ V$. Determine V_o.

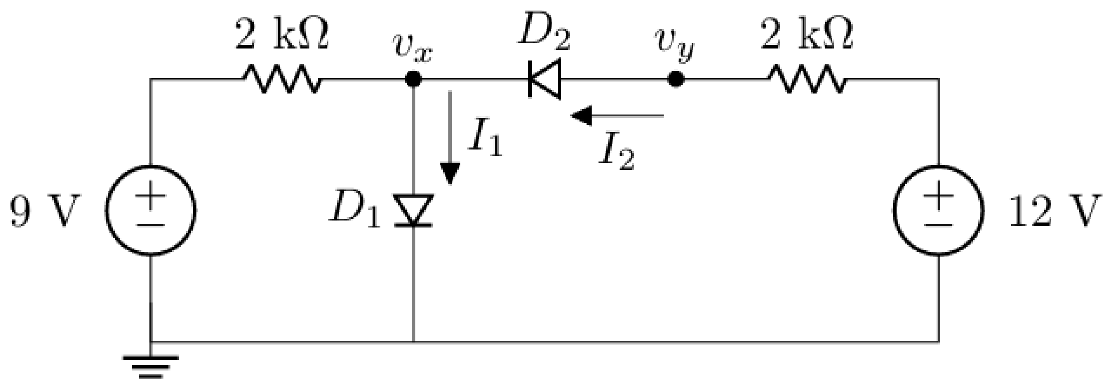


Ans: $D_1 \rightarrow ON, I_{D_1} = 0.16\ mA, V_a = 4.84\ V,\ V_O = V_{Sat}^{+} = 10\ V$

Mid – Spring 2023

- Sherlock Holmes found a piece of paper from the pocket of one of Professor Moriarty's victim with the circuit shown below. One side of the paper is missing and the only other available information was that the D_1 diode is ON. Sherlock needs to know the values of v_x, v_y, I_1, and I_2 as it generates the passcode for the victim's locker, which can help him catch Moriarty. Sherlock knows nothing about diodes and asked for your help to solve the case.

  Calculate the password. Use a cut-in voltage of $V_{D_0} = 1\ V$ for the diodes.

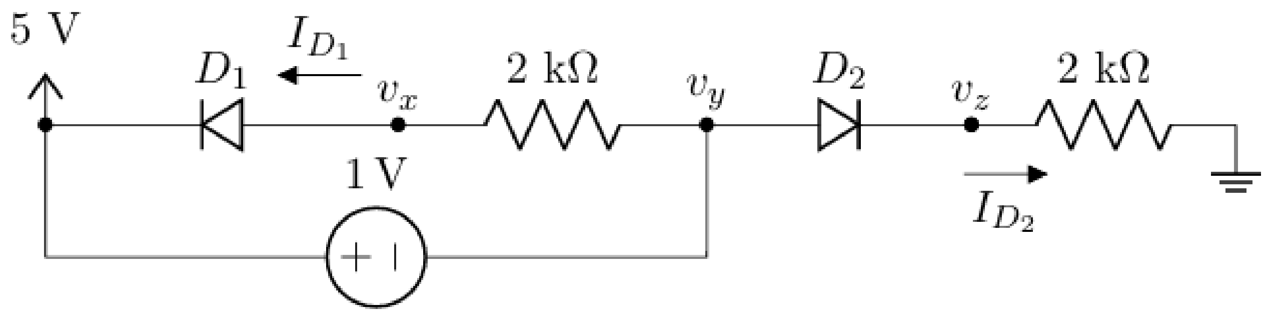


	v_x	I_1	I_2	v_y
Passcode $\Longrightarrow$	?	?	?	?

<u>Ans</u>: $\boldsymbol{D_2 \rightarrow ON, Password = 1\ 9\ 5\ 2}$; All the voltages and currents are in V and mA units, respectively.

Mid – Fall 2022

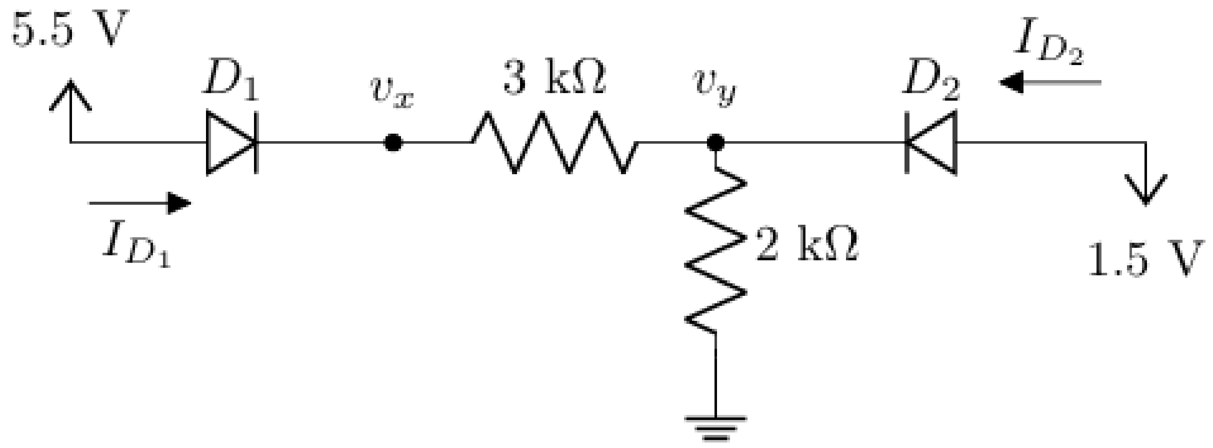
- Using the CVD model with $V_{D_0} = 0.5\ V$ for the diodes, determine I_{D_1}, I_{D_2}, v_x, v_y, and v_z.



Ans: $\boldsymbol{D_1 \rightarrow OFF, D_2 \rightarrow ON, I_{D_1} = 0\ mA, I_{D_2} = 1.75\ mA, v_x = 4\ V, v_y = 4\ V, v_z = 3.5\ V}$

Mid – Summer 2022

- Using the CVD model with $V_{D_0} = 0.5\ V$ for the diodes, determine I_{D_1}, I_{D_2}, v_x, and v_y.



Ans: $D_1 \rightarrow ON, D_2 \rightarrow OFF, I_{D_1} = 1\ mA, I_{D_2} = 0\ mA,\ v_{D_2} = -0.5\ V, v_x = 5\ V, v_y = 2\ V$

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
2. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill